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# Demographic-Based Opinion Mining for Productive Knowledge Sharing

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FIRST
PROJECT REVIEW

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# Problem Statement

**Brief description of the problem:**

- Current group-forming systems rely on basic methods like 'first-come, first-served' and only check resource availability.
- They fail to consider people's interests, intellect, or ideas. We need a smart, secure system that uses opinions and demographics to create productive, like-minded groups.

**Why is it important to solve:**

- Grouping people by shared interests promotes knowledge exchange.
- This exchange directly increases human understanding and thought.
- Grouping similar minds encourages innovative ideas for group betterment.

**Relevance to real-world issues:**

- Unlike fixed surveys, this system uses a smart, learning program.
- It updates people's interests often, about every two weeks.
- This makes sure the groups stay relevant as interests change.

# Objectives

## Clear, measurable goals of the project:

- Make an algorithm that learns by itself to sort people by what they like, using a private cloud.
- Allow the algorithm to learn from what users do and update their interest profiles often.
- Measure how well people match by combining their location with their personal traits, like culture and habits.

## Short-term and long-term outcomes expected:

- Moving away from “first-come, first-served” registration to an “outcome-based” interaction model for mass gatherings.
- Creating a sustainable network of knowledge sharing that improves the human interface across various social and professional sectors.
- Enhancing the “thought process” of society by encouraging innovative ideologies through conceptual-based gatherings.

# Innovation and Uniqueness:

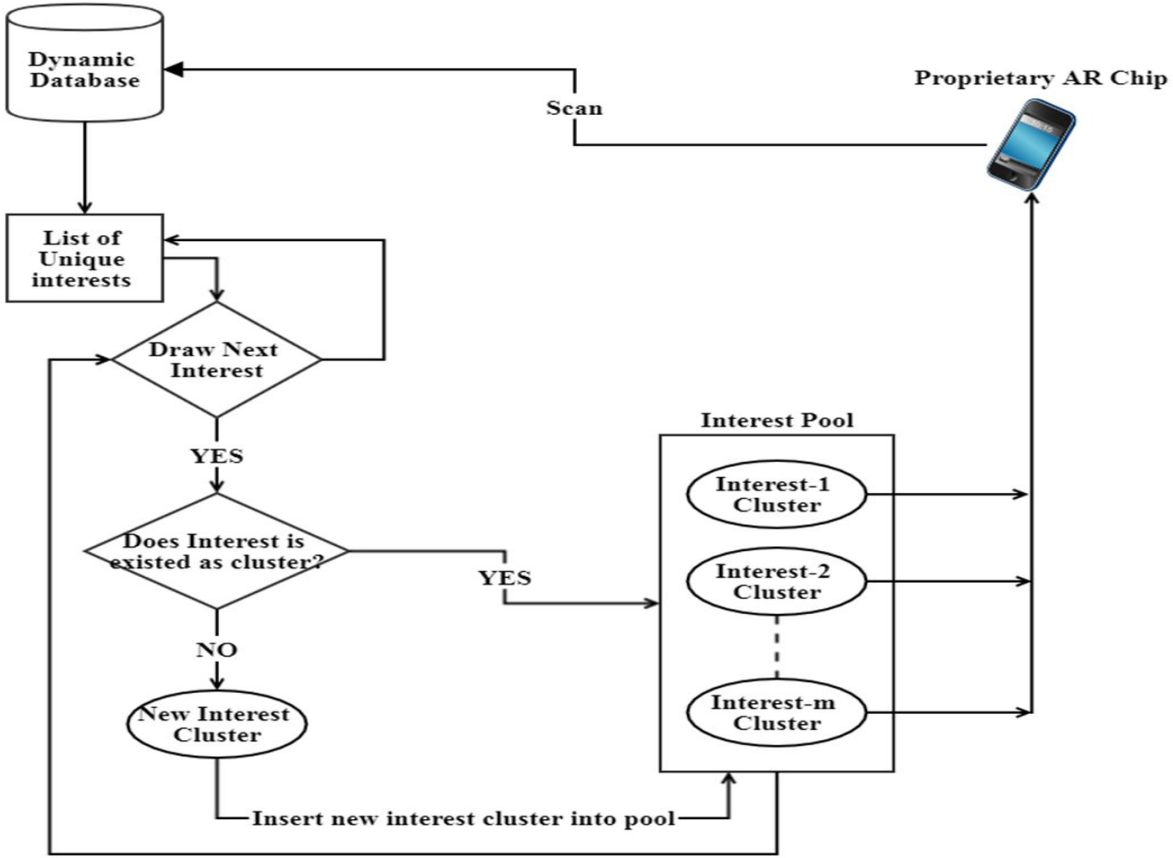
- Unlike existing applications that act as mere resource allocators, this method prioritizes the "intellectual levels" of participants to ensure interactions are productive.
- It uses an exclusive self-learning algorithm in a private cloud to identify similarities in human data that are mostly "unsupervised," meaning the system discovers hidden patterns in behavior rather than relying on pre-labeled categories.
- The innovation involves joining "spatial values" (location) with "profiling parameters" (interests/habits) to calculate the "demographic distance" between individuals in real-time.
- The system doesn't stay static; it updates profiling parameters on a "live" or frequent basis (e.g., every fortnight) to reflect how a person's interests and ideologies change over time.

# Proposed Methodology

- **Phase 1 (Data Gathering & Pruning):** The system collects user interests and ideas, figures out how much support they have, and removes irrelevant data using simple rules.
- **Phase 2 (Cluster Formation in Private Cloud):** Groups of repeating interests are created based on how sure the system is, and these are sorted by a self-learning program to form secure interest groups in a private cloud.
- **Phase 3 (Spatial-Demographic Integration):** User profiles are made by combining their interests with their real-time location (GPS) to show how everyone active is related globally.
- **Phase 4 (Live Social Clustering):** When a user is active, the system calculates their "demographic distance" to nearby people and uses AR scanning to find and group compatible individuals into helpful "category buckets."
- **Phase 5 (Feedback & Self-Learning):** The self-learning program frequently updates the user interest profiles and distance calculations every two weeks based on what the user does to ensure the groups change as the users change.

The methodology is a **Hybrid Unsupervised Clustering Algorithm** that combines **Association Rule Mining (Lattice/Pruning)** with **Spatial-Demographic Distance Mapping**.

# System Architecture / Block Diagram



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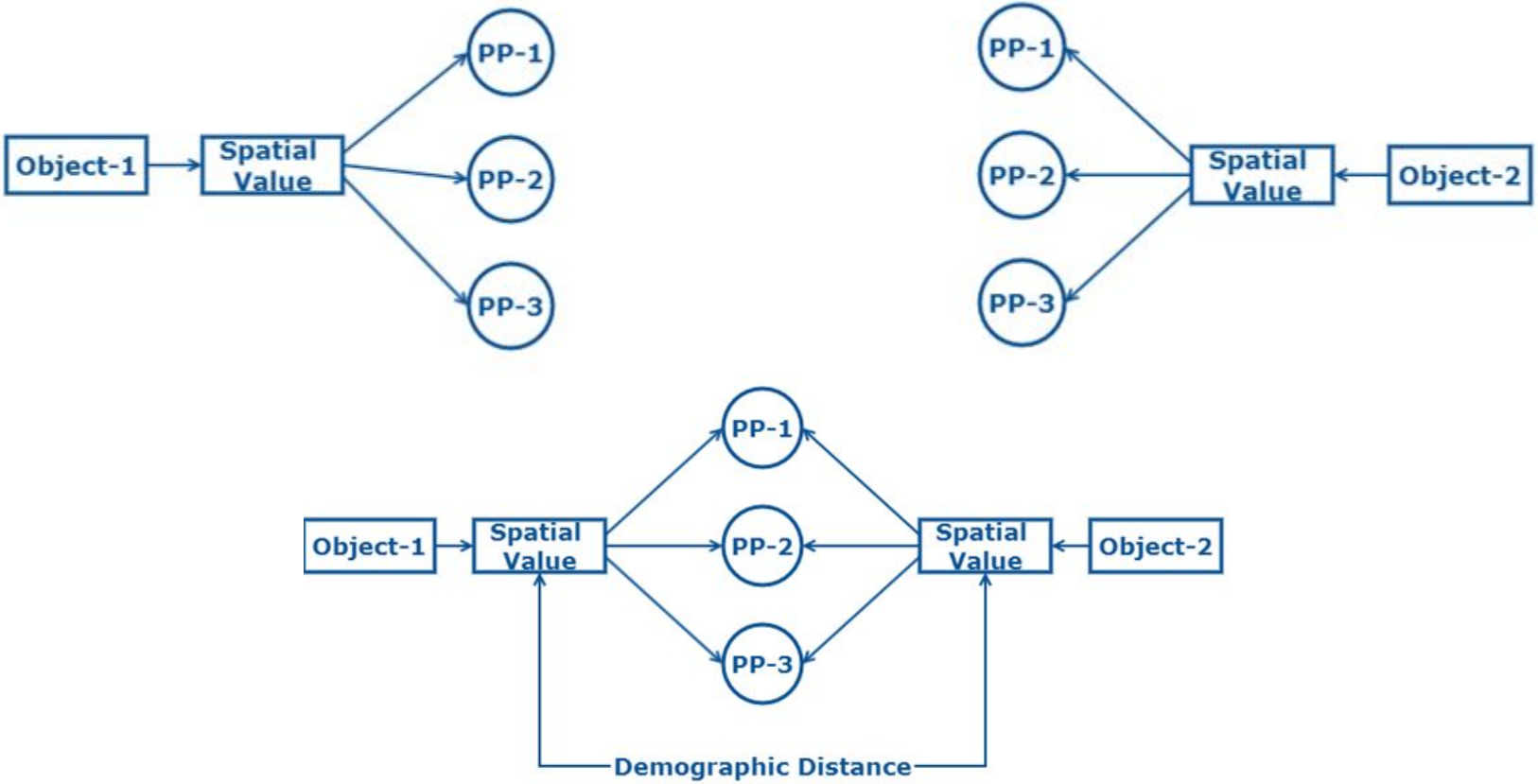

# Expected Outcomes

- Users can immediately identify similar-minded individuals in their physical vicinity using location-based filtering tools.
- Shifts the focus of mass gatherings from simple physical attendance to high-value, productive interactions.
- Creates a sustainable network of knowledge sharing that improves the overall "human interface" in social and professional sectors.
- Increases general productivity in collaborative environments by ensuring team members or event attendees are intellectually aligned.
- The exclusive self-learning algorithm ensures that as people grow and their interests change, their social clusters and "demographic distances" update automatically to stay relevant.

# Work Plan / Timeline

- **Month 1:** Data Gathering & Pruning (Extract user interests and prune low-relevance data).
- **Month 2:** Cluster Formation in Private Cloud (Generate secure interest clusters using unsupervised learning).
- **Month 3:** Spatial-Demographic Integration (Merge user profiles with real-time spatial values to map global relations).
- **Month 4:** Live Social Clustering (Calculate "demographic distances" and use AR scanning to filter and pool compatible individuals).
- **Month 5:** Feedback & Self-Learning (Ongoing recalibration of profiling parameters and demographic distances every fortnight).
- **Month 6:** Documentation and Reference (Final report writing and referencing).

# Resources & Risk Assessment

**Resources and equipment**

- Exclusive self-learning algorithm hosted in a private cloud environment.
- Mobile devices and the necessary software for scanning and location-based filtering tools.
- Secure data storage for profiling parameters and user interactions.

**What are the potential risks (technical/schedule) and how will you mitigate them?**

- **Technical Risk (Algorithm Drift):** The program that learns by itself may not keep up with how people's interests change.
  - **Mitigation:** We will frequently update the interest profiles (like every two weeks) to make sure the system is accurate and relevant.
- **Schedule Risk (Integration Complexity):** It will be complicated to combine real-time location data with the in-depth personal interest profiles.
  - **Mitigation:** We will introduce the location-and-profile combination and the live social grouping in stages to ensure everything works correctly before a full launch.

# Team Roles and Responsibilities

| Role | Responsibilities |
|---|---|
| Project Lead (J3) | Coordinate team progress.<br>Resolve issues.<br>Main point of contact with mentor/advisor. |
| Model Developer (F3) | Design the quality control framework.<br>Develop self-learning algorithm.<br>Integrate software components. |
| Application Developer (D2) | Develops the web application by using the JavaScript. |
| Documentation & Presentation Lead (F4) | Maintain detailed documentation.<br>Prepare slides and reports for the project. |

# Conclusion

- **Summary of strengths of the project**
  - It changes group events from just showing up to having meaningful interactions.
  - The system helps people with similar ideas meet for better work and social time.
  - A special, learning program keeps social groups current as people's interests change.
- **Final remarks or future scope**
  - Creates a lasting network that improves human interaction in various interests.
  - The project is expected to enhance society's thinking by encouraging new ideas through focused interactions.
  - Future plans involve expanding the special program's analysis to cover more social and professional data.

Thank you!